

PilotFish XPath Forking Module – Complete Reference (26R1.11 WAR)

Derived from decompiled eip.war.hs.26R1.11

August 4, 2026

At a glance (plain English)

When one XML file contains **many records** (for example multiple `<Claim>` elements), this fork module **splits it into separate child transactions** — one per node matched by your XPath. Each child carries a copy of the parent's attributes plus a fork ID.

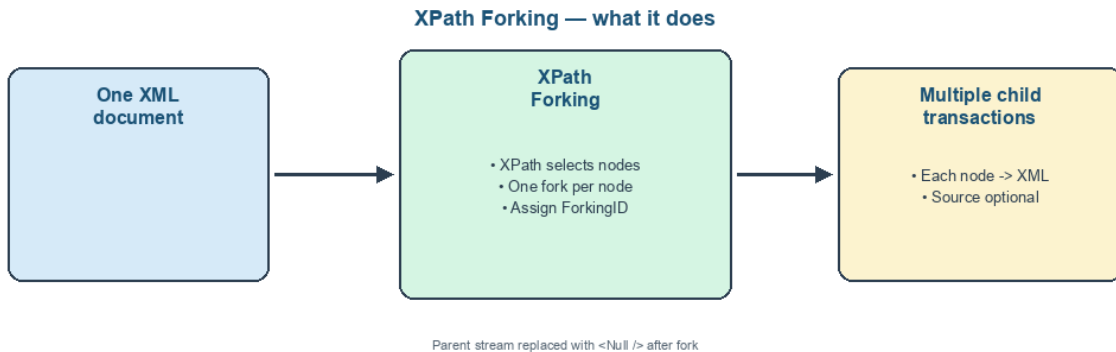


Figure 1: At a glance (plain English)

- **Selects** nodes with an XPath expression (Jaxen / DOM).
- **Emits** one transaction per matched node as standalone XML.
- **Optionally keeps** the original document as an additional fork.
- **Demo / eiConsole type** string is **XPath** (class: XPathForkingModule).

Purpose

Deep dive into **XPath Forking** (XPathForkingModule) in `eip.war.hs.26R1.11`.

Module kind: This class extends **AbstractForkModule** (ModuleType.FORK), not **AbstractProcessor**. It runs in a **fork** stage. Documentation is stored under Processors for parity with the documentation tracker; configure it as a fork module in routes.

Provenance	Value
WAR	<code>eip.war.hs.26R1.11</code>
Module JAR	<code>xcs-modules-core-1.0.3.jar</code>
Implementation class	<code>com.pilotfish.eip.modules.other.XPathForki</code>
Type string	XPath
UI title	XPath Forking Module
Description	Breaks up an XML document based on an XPath expression.
Categories	XML, XPATH

1. Conceptual model

Parent XML transaction

```
|
v
Parse DOM (Xerces DOMParser)
|
v
Jaxen DOMXPath(ForkingXPath) + Namespaces
|
v
For each matched Node:
    wrap node in new Document -> JDOM XMLOutputter -> cache stream
    forkTransaction(child) with ForkingID, ForkingSize, cloned attrs
    |
    v
Optional: fork original bytes (KeepForkingSource)
    |
    v
allForksCompleted(); parent stream -> "<Null />"
```

2. High-level behavior (processTransaction)

1. Set `com.pilotfish.forking.KeepForkingSource` from config **Keep Source Transaction**.
2. Read full payload bytes; parse with `DOMParser` (Xerces).
3. Restore parent stream to `ByteArrayInputStream(copy)` during parse.

4. Build DOMXPath from **ForkingXPath** string.
 5. Apply **Namespaces** table to SimpleNamespaceContext.
 6. `xpath.selectNodes(document)` -> list of nodes.
 7. Log info: module name + node count.
 8. For each node:
 - Build standalone Document (import node if element; use document directly if node type 9).
 - Serialize via JDOM XMLOutputter into EIP cache.
 - New TransactionData with result stream.
 - Copy attributes: `importAttributes(data.getAttributes().cloneAllForResponse())`.
 - Set `com.pilotfish.ForkingID` (int, starting at 1) and `com.pilotfish.ForkingSize` (bytes from `available()`).
 - `forkControl.forkTransaction(newTransaction)`.
 9. If zero nodes and **Exception on No Nodes Returned** true -> `EIPEException("XPath expression returned no nodes")`.
 10. If **Keep Source Transaction** true -> fork another transaction with original byte stream + cloned attrs.
 11. `forkControl.allForksCompleted()`.
 12. Return parent with stream `StringInputStream("<Null />")`.
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3. Configuration reference

3.1 XPath Expression

Tag | `ForkingXPath` |
 Required | yes |
 Tooltip | The XPath expression to fork upon |

3.2 Namespaces

Tag | `Namespaces` |
 Columns | Prefix / URI |

3.3 Exception on No Nodes Returned

Tag | `ExceptionOnNoNodes` |
 Default | `false` |

3.4 Keep Source Transaction

Tag | `KeepForkingSource` |
 Default | `false` |
 Tooltip | Specifies whether or not to pass the original transaction along with the forked transactions |

Also written at runtime to `com.pilotfish.forking.KeepForkingSource` on the parent.

3.5 Fork module base

`AbstractForkModule` adds **no** extra configuration items beyond `AbstractEIPModule` — there is **no** `ExecuteProcessor` flag on fork modules.

4. Transaction attributes

Attribute	Type	Set on
<code>com.pilotfish.ForkingID</code>	incrementing int (stored as object)	each fork child
<code>com.pilotfish.ForkingSize</code>	<code>Integer</code> (stream <code>available()</code>)	each fork child
<code>com.pilotfish.forking.KeepForkingSource</code>	<code>Boolean</code> config mirror	parent before fork

Registered via `registerAttributes()` for `eiConsole`.

5. Worked example

Input:

```
<Batch><Claim id="1"/><Claim id="2"/></Batch>
```

ForkingXPath: `/Batch/Claim`

Result: two child transactions:

- `<Claim id="1"/>` wrapped as document, `ForkingID=1`
- `<Claim id="2"/>` wrapped as document, `ForkingID=2`

Parent ends with `<Null />` body unless **Keep Source Transaction** is enabled.

6. Known quirks and caveats

1. **Jaxen XPath 1.0 style** — not Saxon; differs from **XPath Evaluation** processor.
 2. **Parent stream nulled** — downstream parent-route stages see `<Null />`.
 3. **ForkingSize uses `available()`** — sizing hint, not guaranteed content length.
 4. **Text / attribute nodes** — `getDocumentFromNode` returns null for unsupported node types; may yield empty fork output.
 5. **Attribute clone uses `cloneAllForResponse`** — response-scoped clone semantics apply.
 6. **No matches** — silent unless `Exception` option enabled.
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7. Operational recommendations

Goal	Guidance
Batch XML	Match one repeating element per business record
Parent route	Do not expect meaningful body after fork — process children
Preserve original	Enable Keep Source Transaction for audit/replay forks
Zero matches	Enable exception when empty input must fail
Namespaces	Required for prefixed paths in XPath

8. Source index

Topic	Path
XPath Forking	<code>com/pilotfish/eip/modules/other/XPathForkingModule</code>
AbstractForkModule	<code>com/pilotfish/eip/extend/AbstractForkModule.java</code>
Module JAR	<code>xcs-modules-core-1.0.3.jar</code>

Deep-dive – XPath Forking Module – 26R1.11 – August 2026.